Printed Pages:02 **Sub Code:KEC-403**

Paper Id: 238571 **Roll No.** | | | | | | | | | | |

B-TECH
(SEM IV) THEORY EXAMINATION 2022-23
SIGNAL SYSTEM

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

(a) Determine whether it is energy or power signal $x(t) = tu(t)$

(b) Find even and odd component of the following signal $x(t) = 1 + 2t + 3t^2 + 4t^3$

(c) What is the difference between cross correlation and auto correlation?

(d) Determine the initial value for the given laplace transform

$$X(s) = \frac{4}{s^2 + 3s + 8}$$

(e) What is the significance of ROC

(f) Write time shifting property of Laplace transform

(g) What is the condition of z transform to exist

(h) Draw ROC of two sided sequence in z transform

(i) What is meant by anti-aliasing filter

(j) Find sampling frequency of signal: 2 Cos (2π x 10^3t) Cos (4π x 10^3t)

SECTION B

2. Attempt any *three* of the following: 10x3=30

(a) Check whether the given system is Static, Linear, Causal and Time invariant system or not $y(t) = at^2x(t) + bt\,x(t-4)$

(b) Consider a causal LTI system with frequency response $H(\omega) = \frac{1}{4+j\omega}$.For a particular input x(t) the system is observed to produce the output $y(t) = e^{-2t}u(t) - e^{-4t}u(t)$.Find the input x(t)

(c) Find the Fourier transform of a signal $x(t) = \frac{1}{a^2+t^2}$

(d) Find the Z- transform and ROC of X(z) for

$$x(n) = 3\left(\frac{5}{7}\right)^n u(n) + 2\left(-\frac{1}{3}\right)^n u(n)$$

(e) State and prove sampling theorem

SECTION C

3. Attempt any *one* part of the following: 10x1=10

(a) Sketch the following signal

(i) $2u(t+2) - 2u(t-3)$

(ii) $r(t) - r(t-1) - r(t-3) + r(t-4)$

(b) Check whether the following digital systems are BIBO stable or not

(i) $y(n) = ax(n) + b$

(ii) $y(n) = e^{-x(n)}$

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4. Attempt any *one* part of the following: 10x1=10

(a) Find the convolution of the following signal by graphical method

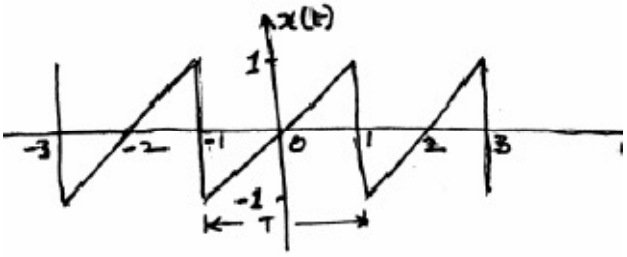
$$x(t) = e^{-3t}u(t); h(t) = u(t+3)$$

(b) Using Fourier transform, find the convolution of the signals

$$x_1(t) = e^{-2t}u(t) \; and \; x_2(t) = e^{-3t}u(t)$$

5. Attempt any *one* part of the following: 10x1=10

(a) Determine the exponential fourier series for the waveform shown



(b) Find the Fourier transform of the signal $x(t) = e^{-a|t|}sgn(t)$

6. Attempt any *one* part of the following: 10x1=10

(a) Find inverse Z-Transform of

$$X(z) = \frac{2z-7}{z^2-5z+6}; ROC; |z| < 2$$

(b) A causal LTI system is described by the difference equation

$$y(n) = y(n-1) + y(n-2) + x(n) + 2x(n-1).$$

Find the system function and frequency response of the system. Plot the poles and zeroes and indicate ROC.Also determine the stability and impulse response of the system

7. Attempt any *one* part of the following: 10x1=10

(a) Solve y[k+2]+ 5y[k+1]+ 6y[k] = 3f [k+1] + 5f[k]
If the initial conditions are y[-1] =11/6, y[-2] = 37/36 and the input f [k] = 2^{-k} V[k] and indicate zero input and zero state component in the out put y[k].

(b) Determine the Nyquist sampling rate and Nyquist sampling interval for

i) $x(t) = sinc(100\pi t) + 3sinc^2(60\pi t)$

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